CENG381 Stochastic Processes

Recurrence Relations - Question Set 1

Key definitions and method summary:

a(n)	the n-th term of a sequence, for n = 0, 1, 2, ...
Homogeneous RR:	a(n) = c1*a(n-1) + c2*a(n-2) + ... + ck*a(n-k) (right-hand side has no standalone function of n)
Non-homogeneous RR:	a(n) = c1*a(n-1) + ... + ck*a(n-k) + f(n) (right-hand side includes an extra term f(n))
Characteristic eqn:	substitute a(n) = r^n into the homogeneous RR to get a polynomial equation in r; its roots r1, r2, ... give the building blocks of the general solution

*General solution: if r1 != r2 (distinct roots), then a(n) = A*r1^n + B*r2^n for constants A, B determined by initial conditions. If r1 = r2 = r (repeated root), then a(n) = (A + B*n)*r^n.*

Q1 (30 points). Solve the following linear homogeneous recurrence relation with the given initial conditions:

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a(n) = 5*a(n-1) - 6*a(n-2)   for n >= 2
Initial conditions:  a(0) = 1,  a(1) = 4
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a) Write down the characteristic equation by substituting a(n) = r^n into the recurrence. Divide through by r^(n-2) to obtain a quadratic in r.

b) Find both roots r1 and r2 of the characteristic equation. Since r1 != r2, write the general solution as a(n) = A*(r1)^n + B*(r2)^n.

c) Apply the initial conditions a(0) = 1 and a(1) = 4 to find the constants A and B, then write the explicit closed-form formula for a(n).

Q2 (35 points). Every unit of time, each living species produces 2 offspring species, and all species that are exactly 2 time units old die. The population starts at time n=0 with 100 newborn species.

a) Let a(n) = total number of species alive at time n. Explain why a(n) = 2*a(n-1) + 2*a(n-2) does NOT correctly model this situation. Then set up the correct recurrence relation. (Hint: species alive at time n = offspring of those alive at n-1 plus those alive at n-1 that are still young enough to survive. Species 2 units old at time n-1 die, so only species aged 0 or 1 at n-1 survive.)

b) The correct recurrence is a(n) = 2*a(n-1) + 2*a(n-2) with a(0) = 100 and a(1) = 200. Solve it using the characteristic equation method.

c) Find the smallest n such that a(n) > 1,000,000 (one million). You may use the approximation that the dominant term in your solution grows as C*(1 + sqrt(3))^n for some constant C.

Q3 (35 points). Solve the following linear NON-homogeneous recurrence relation:

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a(n) = 3*a(n-1) + 2*n   for n >= 1
Initial condition:  a(1) = 3
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*Method: (1) solve the associated homogeneous RR a(n) = 3*a(n-1) to get the homogeneous solution a_h(n); (2) find a particular solution a_p(n) by substituting a trial function of the same form as f(n) = 2*n into the full RR; (3) the general solution is a(n) = a_h(n) + a_p(n); (4) apply the initial condition to fix the free constant.*

a) Solve the homogeneous part a(n) = 3*a(n-1) to get a_h(n) = A * 3^n for some constant A.

b) Find a particular solution a_p(n) of the form a_p(n) = c*n + d by substituting into a(n) = 3*a(n-1) + 2*n and matching coefficients of n and the constant term.

c) Write the general solution a(n) = a_h(n) + a_p(n), apply a(1) = 3 to find A, and give the final closed-form answer.